WANG Can

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Education

The Hong Kong Polytechnic University Sep. 2022 - Oct. 2024
MSc in Urban Informatics and Smart Cities *GPA: 3.03/4.3*

Capital Normal University Sep. 2018 - Jun. 2022
BSc in Geographic Information Science *GPA: 3.3/5*

Research Interests

My research interests lie in multimodal visual understanding and generation, with a focus on grounded reasoning, controllable content creation, and trustworthy generative AI.

Publications

Can Wang, Yuhao Wang, Yushe Cao, Canran Xiao, and Fei Shen. "LaP-Forensics: Latent-Pixel Consistency Guided Multimodal Reasoning for Deepfake Detection." In *Proceedings of the 34th ACM International Conference on Multimedia (MM '26)*, 2026, to appear. doi:10.1145/3767308.3836367. arXiv:2607.25962.

Research Experience

LaP-Forensics: Evidence-Grounded Multimodal Reasoning for Deepfake Analysis 2025 - 2026
Research project; accepted at ACM Multimedia 2026
Methods: Multimodal large language models, DDIM inversion, CLIP, SAM, supervised fine-tuning, GRPO

- **Problem:** Investigated plausible but unsupported artifact explanations in MLLM-based synthetic-image forensics.
- **Method:** Designed a dual-stream model that aligns RGB semantics with DDIM reconstruction residuals; structured Where-What-Why outputs condition SAM-based artifact localization, with SFT followed by GRPO alignment.
- **Result:** On the 246-example curated SynthScars-CoT test split, achieved 72.19% mIoU and 63.62% F1, improving by 9.14 and 8.22 points over the SFT-only baseline, respectively.

Dance-Video Pose Transfer with Wan2.1-VACE Aug. 2025 - Oct. 2025
Industry research project
Methods: Wan2.1-VACE, DiffSynth, LoRA, DeepSpeed

- **Problem:** Targeted limb artifacts, identity drift, and temporal flicker in pose-guided dance-video generation.
- **Method:** Built a 5,000+ video evaluation corpus and fine-tuned Wan2.1-VACE with LoRA.
- **Result:** On a held-out test split, LoRA fine-tuning improved pose AP from 0.58 to 0.67, identity consistency (CSIM) from 0.55 to 0.61, and reduced the limb-dropout rate by 12% over the zero-shot Wan2.1-VACE baseline.

Real-Time 3D Hand Pose Estimation via Asynchronous RGB-Event Fusion Sep. 2024 - Mar. 2025
Research project, DVSense
Methods: RGB-event fusion, deformable alignment, MANO, multi-scale feature modeling

- **Problem:** Targeted two-hand 3D pose estimation under fast motion, low visibility, exposure variation, and hand-hand occlusion.
- **Method:** Built a cached RGB branch for structural features and a fast event-buffer branch for high-frequency motion cues, followed by deformable spatiotemporal alignment and MANO parameter regression.
- **Result:** Reduced MPJPE by 5 mm (52%) and inference latency by 20 ms relative to the single-RGB baseline.

Research Experience (continued)

Building Change Detection on Panoramic Street-View Images Aug. 2023 - Apr. 2024
Master's thesis
Methods: Siamese vision transformer, feature matching, instance-level change detection

- **Problem:** Formulated building-change detection as instance-level visual inconsistency localization under viewpoint shift, occlusion, and pseudo-building distractors.
- **Method:** Constructed a benchmark of 3,000+ image pairs and 10,000+ building instances and developed a Siamese ViT feature-matching pipeline.
- **Result:** Improved recall by 37% and accuracy by 16% over a full-frame change-detection baseline.

Ground-Subsidence Prediction in the Beijing-Tianjin-Hebei Region Sep. 2021 - Apr. 2022
Bachelor's thesis
Methods: T-GCN, LSTM, PS-InSAR, graph-based spatiotemporal modeling

- **Problem:** Modeled regional subsidence forecasting from spatially coupled observations and hydrogeological factors.
- **Method and result:** Built a T-GCN/LSTM forecasting pipeline over PS-InSAR time series, establishing a graph-based model for regional deformation prediction.

Professional Experience

DVSense (Beijing) Technology Co., Ltd. Sep. 2024 - Present
Algorithm Engineer

- Conduct multimodal RGB-event research, including dataset construction, model validation, and robustness evaluation for challenging hand-motion scenes.
- Developed a Qwen3-based retrieval-augmented customer-service system, including knowledge-base construction and response-quality evaluation.

SenseTime May 2024 - Aug. 2024
LLM Evaluation Intern

- Evaluated SenseNova with a five-axis rubric; analyzed instruction-following, factual-consistency, and multi-turn reasoning failures; integrated the model API into OpenCompass for C-Eval.

Otto Poon Fund Institute for Smart Cities, The Hong Kong Polytechnic University Aug. 2022 - Aug. 2023
Research Assistant

- Applied semantic segmentation to land-use category harmonization and developed data-conversion and annotation tools for the Greater Bay Area Big Data Platform.

Honors and Awards

Outstanding Graduate, Capital Normal University 2022
University First-Class Scholarship, Capital Normal University 2021
University Third-Class Scholarship, Capital Normal University 2020

Technical Skills

Computer Vision	Transformers, CNNs, CLIP, SAM, Visual Grounding, Pose Estimation, RGB-Event Fusion
GenAI Related	MLLMs, Diffusion Models, LoRA, SFT, GRPO, RAG
Frameworks	PyTorch, Hugging Face Transformers, TRL, DeepSpeed
Media Production	ComfyUI, Adobe Creative Suite, Video Editing (CapCut, DaVinci)
Programming	Python, C++, LaTeX
Engineering	Linux, Git, Docker
Languages	English (IELTS 7.0), Mandarin (Native), Basic Cantonese and French